

Analysis of a modified fluctuation protocol to estimate plasmid conjugation rates

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ABSTRACT

Accurate estimates of plasmid conjugation rates are required for quantitative studies of horizontal gene transfer. Kosterlitz and colleagues adapted the Luria–Delbrück fluctuation assay to conjugation. Their estimator uses the fraction of replicate cultures containing no transconjugants and therefore does not use the numbers of transconjugants in positive cultures. Here, the same stochastic process is extended to the complete transconjugant-count distribution. Under a deterministic-background approximation, primary donor-to-recipient transfers form a nonhomogeneous Poisson process. Each transfer starts a transconjugant lineage that increases by cell division and secondary transfer to recipients. The endpoint count then has a compound-Poisson probability-generating function. Forward equations give a second description of the process, and its first two moments provide numerical checks. One-dimensional quadrature followed by a general coefficient recursion gives the complete probability mass function. Bernoulli thinning accounts for the plated fraction and colony-establishment probability. The resulting likelihood uses every colony count and reduces to the published zero-class likelihood when the data are recorded only as absent or present and detection is complete. Empirical probability-generating-function fitting and Fourier inversion provide alternative computational methods. A joint recipient–transconjugant process is also derived for finite recipient populations. Its marginal transconjugant probability-generating function is not closed, which identifies the limit of the independent-lineage construction and motivates numerical evaluation. These results combine the conjugation model of Kosterlitz and colleagues with the full-count methods of Gerrish and of Vázquez-Mendoza and colleagues.

KEYWORDS conjugation; fluctuation analysis; horizontal gene transfer; Luria–Delbrück distribution; plasmid; probability-generating function

Introduction

Horizontal gene transfer contributes to bacterial adaptation by moving genes among cells that are not related by direct descent (Thomas and Nielsen 2005; Norman *et al.* 2009; Redondo-Salvo *et al.* 2020). Conjugative plasmids are important agents of this process. They carry genes for their own transfer and may also carry traits that affect bacterial survival, including antimicrobial resistance (San Millan 2018; Antimicrobial Resistance Collaborators 2022). Quantitative studies of plasmid ecology therefore

require an estimate of the conjugation rate under a specified set of donor, recipient, plasmid, and environmental conditions.

The standard experimental design mixes plasmid-bearing donors with plasmid-free recipients and measures donor, recipient, and transconjugant densities after a period of growth. Classical estimators are based on deterministic mass-action models (Levin *et al.* 1979; Simonsen *et al.* 1990). Their accuracy can be affected by unequal growth rates, unequal transfer rates from donors and transconjugants, resource limitation, and low transconjugant numbers (Huisman *et al.* 2022; Kosterlitz *et al.* 2022). These problems are most severe when the first few transfer events have a large effect on the final transconjugant population.

Kosterlitz *et al.* (2022) addressed this problem by treating transconjugant production as a stochastic process. Their Luria–Delbrück method estimates the primary donor conjugation

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tion rate from the fraction of independent cultures that contain no transconjugants. The method follows the zero-class argument of [Luria and Delbrück \(1943\)](#): the probability of observing no event depends on the integrated rate at which the first event occurs, but not on the later expansion of the resulting lineage. This property allows the primary conjugation rate to be estimated even when transconjugants and donors have different transfer rates.

The zero class is useful because it is simple and relatively insensitive to the distribution among positive counts. It also discards information. A culture with 1 transconjugant and a culture with 1,000 transconjugants have the same binary score. In mutation-rate estimation, the complete Luria–Delbrück distribution has long been used to distinguish late events from early events that produce large clones ([Lea and Coulson 1949](#); [Zheng 1999](#)). Methods for computing this distribution include coefficient recursions based on the logarithm of its probability-generating function ([Ma et al. 1991](#); [Sarkar et al. 1992](#); [Zheng 2005](#)).

The treatment of sampling is also relevant. Only a fraction of a culture is usually plated, and only a fraction of plated cells may form colonies ([Stewart 1991](#); [Jones et al. 1994](#); [Zheng 2008](#)). [Gerrish \(2008\)](#) composed the Luria–Delbrück probability-generating function with the probability-generating function of Bernoulli sampling and derived an efficient recursion for the resulting counts. This calculation made full likelihood analysis compatible with the extended Jones protocol, in which cultures are grown to a large size and a known fraction is plated.

More recently, [Vázquez-Mendoza et al. \(2024\)](#) considered a fluctuation model whose probability-generating function did not lead directly to a convenient closed-form probability mass function. They fitted the theoretical function to the empirical probability-generating function and also recovered count probabilities by Fourier inversion. These methods separate the biological derivation of a probability-generating function from the numerical calculation used for inference ([Nakamura and Pérez-Abreu 1993](#); [Lange 1982](#)).

Here these developments are applied to the stochastic conjugation model of [Kosterlitz et al. \(2022\)](#). Two related models are considered. The first retains deterministic donor and recipient trajectories and negligible recipient depletion. It gives an analytic compound-Poisson probability-generating function and a practical full-count likelihood. The second retains the finite recipient population in a joint stochastic process. Its marginal transconjugant distribution is not closed, but its joint probability-generating-function equation specifies the calculation required for numerical inference. In the first model, the general coefficient recursion remains available because its coefficients can be evaluated by numerical integration; a closed form is not required.

The conjugation process

Consider a well-mixed culture of volume V . Let $D(s)$ and $R(s)$ be donor and recipient densities at time s , and let $T(s)$ be the number of transconjugants. Thus, D and R have units of cells per unit volume, whereas T is a cell count. The assay starts with $T(0) = 0$. As in the basic model of [Kosterlitz et al. \(2022\)](#), assume deterministic exponential growth during the assay,

$$D(s) = D_0 e^{\psi_D s}, \quad R(s) = R_0 e^{\psi_R s}, \quad (1)$$

where ψ_D and ψ_R are per-capita growth rates. Recipient loss through transfer is neglected. The primary transfer rate from donors to recipients, expressed as a total rate for the culture, is

$$a(s) = V \gamma_D D(s) R(s), \quad (2)$$

where γ_D is the donor conjugation-rate constant.

Each transconjugant produces 1 additional transconjugant either by cell division or by transfer to a recipient. Its per-capita production rate is therefore

$$b(s) = \psi_T + \gamma_T R(s), \quad (3)$$

where ψ_T is the transconjugant growth rate and γ_T is the secondary conjugation-rate constant. In an interval ds , the transition rate from T to $T + 1$ is

$$a(s) + b(s)T. \quad (4)$$

Thus, primary transfer is time-dependent immigration and transconjugant increase is a time-dependent linear birth process.

Let

$$p_n(s) = \Pr\{T(s) = n\}, \quad n = 0, 1, 2, \dots \quad (5)$$

The forward equations are

$$\begin{aligned} \frac{dp_0(s)}{ds} &= -a(s)p_0(s), \\ \frac{dp_n(s)}{ds} &= [a(s) + (n-1)b(s)]p_{n-1}(s) \\ &\quad - [a(s) + nb(s)]p_n(s), \quad n \geq 1, \end{aligned} \quad (6)$$

with $p_0(0) = 1$ and $p_n(0) = 0$ for $n \geq 1$. These equations give a direct numerical calculation after truncation at a sufficiently large count. They also give the probability-generating-function equation. With

$$G(z, s) = \mathbb{E}[z^{T(s)}] = \sum_{n=0}^{\infty} p_n(s)z^n, \quad (8)$$

one obtains

$$\frac{\partial G}{\partial s} = (z-1) \left[a(s)G + b(s)z \frac{\partial G}{\partial z} \right], \quad G(z, 0) = 1. \quad (9)$$

The first term represents primary transfer. The second represents the increase of existing transconjugant lineages by division and secondary transfer.

Figure 1 summarizes this process and shows how primary transfer events seed the independent transconjugant lineages used below.

The complete transconjugant distribution

Suppose that a primary transfer occurs at time s and starts a lineage with 1 transconjugant. Define

$$B(s, t) = \int_s^t b(u) du, \quad q_s = e^{-B(s, t)}. \quad (10)$$

The size $X_{s,t}$ of this lineage at assay time t has the geometric distribution

$$\Pr(X_{s,t} = k) = q_s(1 - q_s)^{k-1}, \quad k \geq 1, \quad (11)$$

with probability-generating function

$$g_{s,t}(z) = \frac{q_s z}{1 - (1 - q_s)z}. \quad (12)$$

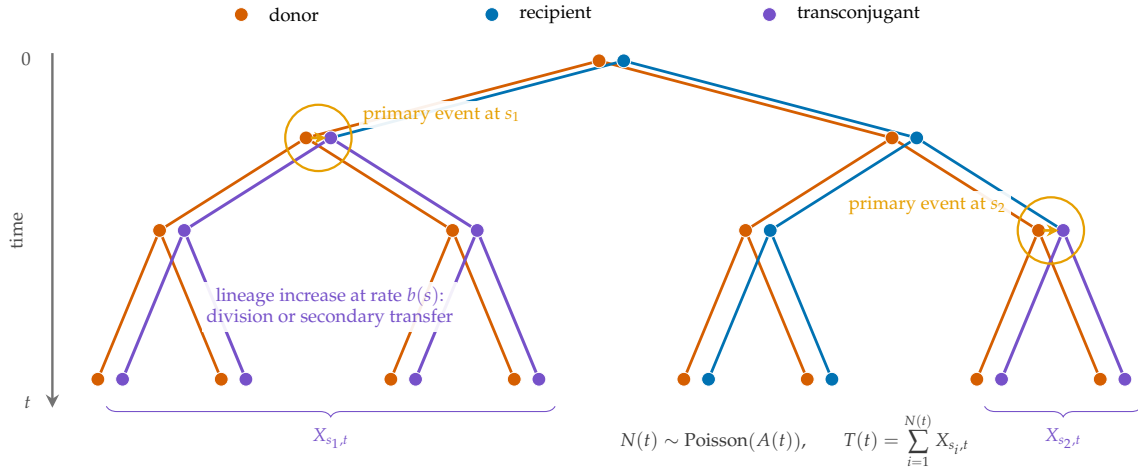


Figure 1 Lineage interpretation of the stochastic conjugation model. Time runs downward from mixing at time 0 to the assay endpoint t . Orange and blue branches represent donor and recipient proliferation. At a primary transfer time s_i (orange arrow), a donor transfers the plasmid to a recipient; the converted cell and the transconjugants descended from that initiation event are purple. The earlier event in this schematic produces a larger terminal lineage, $X_{s_1,t}$, than the later event, $X_{s_2,t}$. In the model, primary events form an inhomogeneous Poisson process with intensity $a(s) = V\gamma_D D(s)R(s)$ and cumulative intensity $A(t)$. Each event seeds an independent lineage whose increases, through cell division or secondary transfer to a recipient, occur at per-capita rate $b(s) = \psi_T + \gamma_T R(s)$. Consequently, the endpoint count is the compound-Poisson sum $T(t) = \sum_{i=1}^{N(t)} X_{s_i,t}$. Branching times and clone sizes are illustrative rather than literal; donor and recipient trajectories are treated as deterministic, and recipient depletion is neglected.

For the exponential recipient trajectory in Equation (1),

$$B(s, t) = \psi_T(t - s) + \frac{\gamma_T R_0}{\psi_R} (e^{\psi_R t} - e^{\psi_R s}), \quad (13)$$

with the continuous limit $\gamma_T R_0(t - s)$ for the second term when $\psi_R = 0$.

Primary transfers form an inhomogeneous Poisson process with rate $a(s)$. Conditional on their initiation times, their descendant lineages are independent. The probability-generating function of the total transconjugant count is therefore

$$G_t(z) = \exp \left\{ \int_0^t a(s) [g_{s,t}(z) - 1] ds \right\}. \quad (14)$$

This is a time-dependent compound-Poisson distribution. A related closed-form probability-generating function was considered in the supporting analysis of Kosterlitz *et al.* (2022) for a restricted case, using results of Keller and Antal (2015). Equation (14) retains secondary conjugation through the time-dependent birth rate in Equation (3).

Define

$$A(t) = \int_0^t a(s) ds, \quad (15)$$

$$\lambda_k(t) = \int_0^t a(s) q_s (1 - q_s)^{k-1} ds, \quad k \geq 1. \quad (16)$$

The value λ_k is the expected number of primary transfers whose lineages contain exactly k cells at time t . Because $\sum_{k \geq 1} \lambda_k = A$, Equation (14) can be written

$$\log G_t(z) = -A + \sum_{k=1}^{\infty} \lambda_k z^k. \quad (17)$$

Let $p_n = \Pr[T(t) = n]$. Differentiating Equation (17) and equating coefficients gives the general recursion (Ma *et al.* 1991; Sarkar *et al.* 1992)

$$p_0 = e^{-A}, \quad p_n = \frac{1}{n} \sum_{k=1}^n k \lambda_k p_{n-k}. \quad (18)$$

The λ_k need not have closed forms. They can be calculated by one-dimensional quadrature. Probabilities through an observed maximum $n_{\max}$ require only $\lambda_1, \dots, \lambda_{n_{\max}}$.

The zero class follows immediately from Equation (15). Under Equation (1),

$$p_0 = \exp \left[-V\gamma_D D_0 R_0 \frac{e^{(\psi_D + \psi_R)t} - 1}{\psi_D + \psi_R} \right], \quad (19)$$

which is the estimator-defining probability in Kosterlitz *et al.* (2022). The published method is therefore the $n = 0$ part of the complete distribution. The continuous limit of Equation (19) applies when $\psi_D + \psi_R = 0$.

Moments and a constant-rate limit

The first two moments follow from the compound-Poisson representation. A lineage initiated at s has first raw moment $e^{B(s,t)}$ and second raw moment $2e^{2B(s,t)} - e^{B(s,t)}$. Consequently,

$$\mathbb{E}[T(t)] = \int_0^t a(s) e^{B(s,t)} ds, \quad (20)$$

$$\text{Var}[T(t)] = \int_0^t a(s) [2e^{2B(s,t)} - e^{B(s,t)}] ds. \quad (21)$$

These expressions provide checks on the recursion, Fourier inversion, and stochastic simulation. If $m(t) = \mathbb{E}[T(t)]$, then

$$\frac{dm(t)}{dt} = a(t) + b(t)m(t), \quad m(0) = 0. \quad (22)$$

Under the density convention used here, this equation is

$$\frac{dm(t)}{dt} = V\gamma_D D(t)R(t) + [\psi_T + \gamma_T R(t)]m(t). \quad (23)$$

Equivalently, $m(t)/V$ satisfies the usual deterministic mass-action equation for transconjugant density. This agreement provides a direct connection between the stochastic count model and the corresponding mean-field model.

For an additional check, suppose that $a(s) = a$ and $b(s) = b > 0$ are constant. Direct integration of Equation (14) gives

$$G_t(z) = \left[z + (1-z)e^{bt} \right]^{-a/b} = \left[\frac{e^{-bt}}{1 - (1 - e^{-bt})z} \right]^{a/b}. \quad (24)$$

Thus, $T(t)$ has a negative-binomial distribution with shape a/b and success probability e^{-bt} . Its mean and variance reduce to $(a/b)(e^{bt} - 1)$ and $(a/b)(e^{2bt} - e^{bt})$, respectively. As $b \rightarrow 0$, Equation (24) converges to the Poisson probability-generating function $\exp\{at(z - 1)\}$.

Plating and colony establishment

Let ρ be the effective probability that a transconjugant present at time t produces a counted colony. It includes the plated fraction and colony-establishment probability. Under independent sampling,

$$C | T \sim \text{Binomial}(T, \rho), \quad (25)$$

and the probability-generating function of the colony count C is

$$H_t(z) = G_t(1 - \rho + \rho z). \quad (26)$$

This is the same probability-generating-function composition used to represent dilution and plating in fluctuation assays (Stewart 1991; Jones *et al.* 1994; Gerrish 2008; Zheng 2008; Vázquez-Mendoza *et al.* 2024).

The composition can again be put in a form suitable for Equation (18). Set

$$d_s = q_s + (1 - q_s)\rho. \quad (27)$$

A lineage initiated at s contributes no observed colony with probability

$$\Pr(Y_s = 0) = \frac{q_s(1 - \rho)}{d_s}, \quad (28)$$

and contributes $j \geq 1$ colonies with probability

$$\Pr(Y_s = j) = \frac{q_s \rho}{d_s^2} \left[\frac{(1 - q_s)\rho}{d_s} \right]^{j-1}. \quad (29)$$

Define

$$\tilde{\lambda}_j = \int_0^t a(s) \frac{q_s \rho}{d_s^2} \left[\frac{(1 - q_s)\rho}{d_s} \right]^{j-1} ds, \quad (30)$$

$$\tilde{A} = \int_0^t a(s) \frac{\rho}{d_s} ds. \quad (31)$$

Then $\sum_{j \geq 1} \tilde{\lambda}_j = \tilde{A}$, and the observed-count probabilities $h_n = \Pr(C = n)$ satisfy

$$h_0 = e^{-\tilde{A}}, \quad h_n = \frac{1}{n} \sum_{j=1}^n j \tilde{\lambda}_j h_{n-j}. \quad (32)$$

When $\rho < 1$, an observed zero can result either from no primary transfer or from failure to sample the descendants of a transfer. This distinction must be included if a plated zero class is used. Equation (19) is recovered when $\rho = 1$ and transconjugant lineages do not become extinct.

Inference from colony counts

For independent cultures with observed counts $c_1, \dots, c_m$, the full log likelihood is

$$\ell(\theta) = \sum_{i=1}^m \log h_{c_i}(\theta), \quad (33)$$

where θ contains γ_D and any parameters not fixed by separate measurements. If D , R , ψ_T , γ_T , and ρ are fixed, each coefficient in Equation (30) is proportional to γ_D . The required integrals can then be calculated once apart from this common factor, and one-dimensional likelihood maximization is fast.

Equation (33) uses the complete set of observed counts. The binary likelihood used by the zero-class method is obtained by replacing each c_i with $I(c_i > 0)$. Because this replacement is a deterministic reduction of the data, the count experiment cannot contain less Fisher information about a correctly specified parameter than its binary reduction. The gain may be small when nearly every positive culture contains a single detected lineage. It can be substantial when the positive counts vary in ways that distinguish the number and timing of transfer events. As in conventional fluctuation analysis, this gain is accompanied by greater sensitivity to assumptions about lineage growth and sampling (Gerrish 2008; Ycart 2013).

Empirical probability-generating-function estimation

The empirical probability-generating function is

$$\hat{H}_m(z) = \frac{1}{m} \sum_{i=1}^m z^{c_i}. \quad (34)$$

Following Vázquez-Mendoza *et al.* (2024), parameters may be estimated by minimizing the distance between Equation (34) and the theoretical function in Equation (26) at several real values of z between 0 and 1. The original zero class is retained because $\hat{H}_m(0)$ is the observed fraction of zero counts. Values $z > 0$ add information from positive cultures while limiting the effect of very large counts.

For real evaluation points z_j and z_k , the model covariance is

$$\text{Cov} [\hat{H}_m(z_j), \hat{H}_m(z_k)] = \frac{H_t(z_j z_k) - H_t(z_j)H_t(z_k)}{m}. \quad (35)$$

This covariance permits weighted minimum-distance estimation. The evaluation points should be selected by simulation under the intended experimental range rather than copied from another assay. Confidence intervals can be obtained by resampling whole cultures or by parametric simulation.

Fourier inversion

Count probabilities may also be obtained by evaluating the probability-generating function on a complex circle and applying a discrete inverse Fourier transform (Lange 1982; Vázquez-Mendoza *et al.* 2024). A radially scaled form is

$$h_n \simeq \frac{1}{Mr^n} \sum_{\ell=0}^{M-1} H_\ell \left(r e^{2\pi i \ell / M} \right) e^{-2\pi i n \ell / M}, \quad 0 < r < 1. \quad (36)$$

The choice $r < 1$ reduces wrap-around error from the infinite right tail. Values of M and r should be varied until all likelihood contributions for the observed counts are stable. Under the present model, Equation (32) is simpler. Fourier inversion remains useful as an independent numerical check and for extensions in which clone probabilities are obtained numerically.

Recipient depletion and the joint process

The compound-Poisson representation treats donor and recipient trajectories as deterministic and does not remove a recipient after transfer. A finite-recipient model must retain the recipient count in the state. Let $N_R(t)$ and $N_T(t)$ denote recipient and transconjugant counts, respectively, and continue to treat $D(t)$ as a deterministic donor density. Conditional on $D(t)$, the transitions are

$$\begin{aligned} (N_R, N_T) &\longrightarrow (N_R + 1, N_T) && \text{at rate } \psi_R N_R, \\ (N_R, N_T) &\longrightarrow (N_R, N_T + 1) && \text{at rate } \psi_T N_T, \\ (N_R, N_T) &\longrightarrow (N_R - 1, N_T + 1) && \text{at rate } \gamma_D D(t) N_R, \\ (N_R, N_T) &\longrightarrow (N_R - 1, N_T + 1) && \text{at rate } \frac{\gamma_T}{V} N_R N_T. \end{aligned} \quad (37)$$

The third transition is primary transfer, and the fourth is secondary transfer. The factor $1/V$ in the fourth rate is required because both N_R and N_T are counts. It is absent from the third rate because $D(t)$ is a density.

Define the joint probability-generating function

$$F(u, z, t) = \mathbb{E} \left[u^{N_R(t)} z^{N_T(t)} \right]. \quad (38)$$

Application of the Markov-chain generator to $u^{N_R} z^{N_T}$ gives

$$\begin{aligned} \frac{\partial F}{\partial t} &= \psi_R u(u-1) \frac{\partial F}{\partial u} + \psi_T z(z-1) \frac{\partial F}{\partial z} \\ &+ \gamma_D D(t)(z-u) \frac{\partial F}{\partial u} + \frac{\gamma_T}{V} z(z-u) \frac{\partial^2 F}{\partial u \partial z}. \end{aligned} \quad (39)$$

For a fixed initial recipient count $N_{R,0}$ and no initial transconjugants, $F(u, z, 0) = u^{N_{R,0}}$. The marginal transconjugant probability-generating function is $G(z, t) = F(1, z, t)$. Setting $u = 1$ in Equation (39) gives

$$\begin{aligned} \frac{\partial G}{\partial t} &= \psi_T z(z-1) \frac{\partial G}{\partial z} + \gamma_D D(t)(z-1) F_u(1, z, t) \\ &+ \frac{\gamma_T}{V} z(z-1) F_{uz}(1, z, t). \end{aligned} \quad (40)$$

Equation (40) is not closed in G because it depends on the joint state of recipients and transconjugants. Recipient depletion couples lineages that were independent in Equation (14). The complete distribution can instead be calculated by truncating the joint master equation, by solving Equation (39) numerically, or by stochastic simulation. A theoretical probability-generating

function obtained by any of these methods can be used in empirical probability-generating-function fitting or Fourier inversion. If donor numbers are also stochastic, donor count must be added as a third state variable.

The true zero class remains independent of γ_T because secondary transfer cannot occur before the first transconjugant appears. However, when recipient growth is stochastic, its probability is an average over recipient trajectories rather than the deterministic expression in Equation (19). This distinction does not affect Equation (19) under the deterministic-background approximation, but it is relevant when the joint process is fitted directly.

Experimental design and model checks

The binary selective outgrowth used by Kosterlitz *et al.* (2022) should be replaced by direct count measurement. Independent donor-recipient cultures are stopped at a specified time, further growth and conjugation are stopped, and a known fraction of each culture is plated on transconjugant-selective medium. The numerical colony count, including zero, is recorded for every culture. Selective conditions must prevent new conjugation on the plate; this must be checked for each system because post-plating transfer can increase apparent transconjugant counts (Smit and van Elsas 1990; Kosterlitz *et al.* 2022).

Selection for the plasmid alone does not distinguish a transconjugant from a donor because both carry the plasmid. The recipient background therefore requires a second selectable marker or another stable identifying trait that is absent from the donor. Transconjugants are counted under simultaneous selection for the plasmid marker and the recipient marker. Donor-only and recipient-only controls must show that neither parental strain forms colonies under this double selection. Separate non-selective or single-selective plates can be used to estimate donor and recipient densities. The marker combination and the order of stopping, dilution, and plating should be validated for each strain-plasmid pair.

The effective sampling probability ρ should be measured with transconjugant controls. It may differ from the volume fraction because of cell death, incomplete colony establishment, or cell clumping. If several disjoint aliquots are plated, their counts can be summed and the total plated fraction used. If a dilution is selected after its count is observed, the selection rule should be included in the likelihood; a prespecified dilution or a prespecified set of plates is simpler. Counts above a plate-reading limit should be retained as right-censored observations. For a threshold L , their likelihood contribution is

$$\Pr(C \geq L) = 1 - \sum_{n=0}^{L-1} h_n. \quad (41)$$

The extended Jones principle of using a large culture followed by dilution can increase information while keeping colony counts manageable (Jones *et al.* 1994; Gerrish 2008). In a conjugation assay, assay duration cannot be increased without limit. Recipient depletion, stationary phase, density-dependent transfer, and secondary conjugation may violate Equations (1)–(4). The culture volume, stopping time, and plated fraction should therefore be selected by simulation so that counts are measurable while transconjugants remain rare enough for the Kosterlitz assumptions to hold.

The primary rate γ_D controls lineage initiation, whereas ψ_T and γ_T control lineage size. These parameters may be weakly

identified from a single endpoint. Separate measurements of transconjugant growth and secondary conjugation, as used by [Kosterlitz *et al.* \(2022\)](#), reduce this problem. Multiple stopping times or recipient densities provide additional separation. Uncertainty in independently estimated rates and in ρ should be propagated by profile likelihood, parametric bootstrap, or joint inference.

Model checking should include the zero fraction, the empirical probability-generating function, the central count distribution, and the upper tail. Agreement of $\hat{\gamma}_D$ across stopping times and plating fractions provides a direct check of the growth and sampling assumptions. Failure of these checks indicates that a more detailed process is required.

Discussion

The complete count theory combines 3 existing developments. [Kosterlitz *et al.* \(2022\)](#) specified the conjugation process and identified a zero class that isolates primary transfer under their assumptions. [Gerrish \(2008\)](#) showed that plating can be represented by probability-generating-function composition and that full likelihood inference remains practical after dilution. [Vázquez-Mendoza *et al.* \(2024\)](#) showed how empirical probability-generating-function fitting and Fourier inversion can support inference when direct probability calculations are difficult. The combination yields a count-based extension of the conjugation fluctuation assay.

Under deterministic donor and recipient trajectories and negligible recipient depletion, the process is a time-inhomogeneous linear birth process with immigration. Equivalently, it is compound Poisson: each primary transfer starts an independent transconjugant lineage, and the endpoint count is the sum of the lineage sizes. This structure places the logarithm of the probability-generating function in the form required by the Ma-Sandri-Sarkar recursion. The coefficients are one-dimensional numerical integrals rather than closed expressions, but the recursion is unchanged. The forward equations, probability-generating-function equation, and moment formulas give independent descriptions of the same process. The negative-binomial constant-rate case gives a further check.

The assumptions that make the compound-Poisson calculation possible also define its range of validity. The representation is exact conditional on the specified deterministic background trajectories, negligible recipient depletion, pure-birth transconjugant lineages, and independent sampling. It is not an exact representation of a finite culture when realized transfers materially change the recipient population. In the finite-recipient process, primary and secondary transfer both convert recipients into transconjugants, and the marginal transconjugant probability-generating function is coupled to joint recipient-transconjugant quantities. The joint equation derived here gives a formal starting point for a master-equation calculation or stochastic simulation. The methods of [Vázquez-Mendoza *et al.* \(2024\)](#) can then be applied after the theoretical probability-generating function or its numerical approximation has been obtained.

The independent-lineage representation can also accommodate several changes without introducing recipient coupling. If a lineage has probability-generating function $g_{s,t}(z)$ under a model with death, plasmid loss, incomplete establishment, or a lag period, Equation (14) remains valid as long as primary events have deterministic intensity and their lineages remain independent. Lineage extinction then contributes to the true zero probability, as considered by [Kosterlitz *et al.* \(2022\)](#) and in stochastic

establishment models ([Alexander and MacLean 2020](#)). The logarithmic coefficients can be obtained from a numerical lineage distribution or from evaluations of the probability-generating function.

Full counts add information about the number and timing of primary events and about subsequent lineage increase. They also make inference more dependent on the models for transconjugant growth, secondary transfer, recipient availability, and sampling. For this reason, the binary and full-count analyses should both be reported. Agreement of their estimates over assay durations, recipient densities, and plated fractions would support the model; systematic disagreement would identify conditions under which a joint process or an additional biological mechanism is required.

Data availability

No empirical data were generated or analyzed in this study. The method is defined by the equations in the article.

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Conflicts of interest

The author declares no conflict of interest.

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